Math 421

Wednesday, November 26

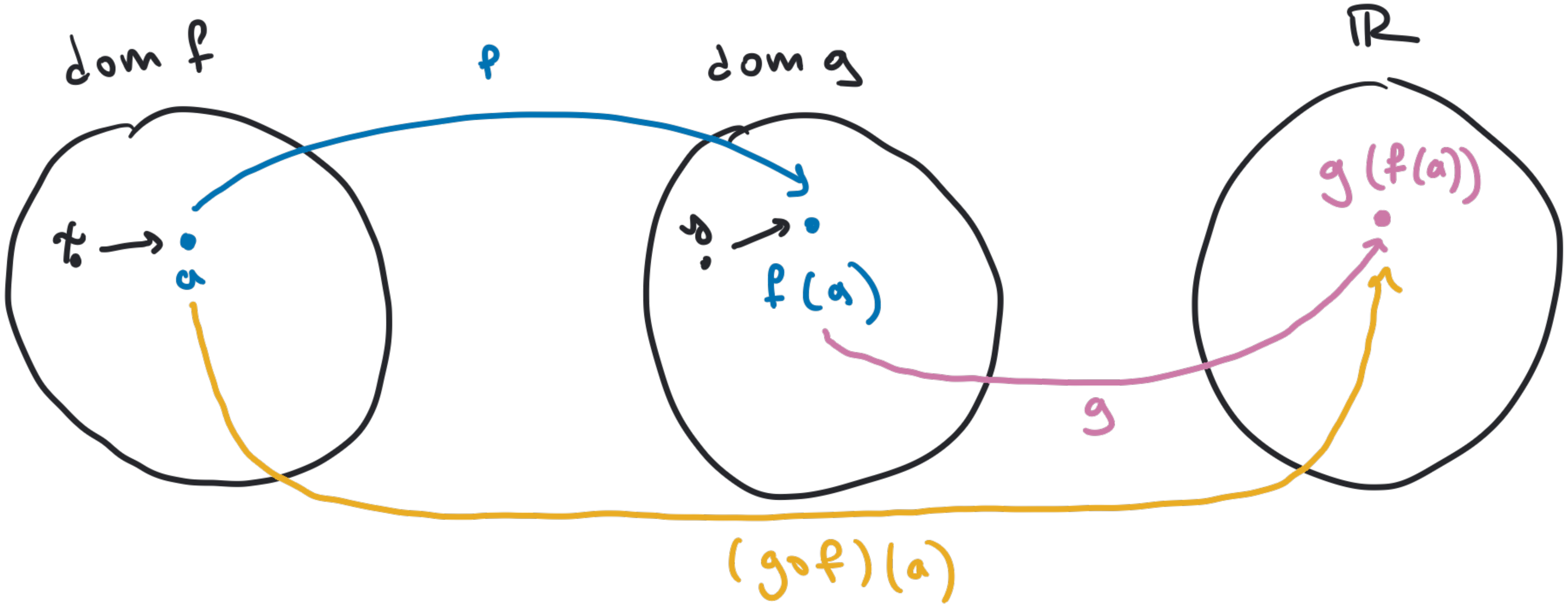
Announcements

- This week's drop-in hours:
  - **Today** 12-1 pm VV 311
- Check for any final exam conflicts & email me
- Have a great break!

Chain rule

Theorem 28.4. If f is differentiable at a and g is differentiable at $f(a)$, then $g \circ f$ is differentiable at a and

$$(g \circ f)'(a) = g'\big(f(a)\big) \cdot f'(a)$$



$$\mathrm{dom}(g\circ f) = \{x \in \mathrm{dom}\, f : f(x) \in \mathrm{dom}\, g\}$$

"Proof" of Chain Rule

$$\lim_{x\to a} \frac{(g\circ f)(x) - (g\circ f)(a)}{x-a} = \lim_{x\to a} \frac{g(f(x)) - g(f(a))}{x-a}$$

$$= \lim_{x\to a} \frac{g(f(x)) - g(f(a))}{x-a} \cdot \frac{f(x)-f(a)}{\boxed{f(x)-f(a)}}$$

$$= \lim_{x\to a} \frac{g(f(x)) - g(f(a))}{f(x)-f(a)} \cdot \frac{f(x)-f(a)}{x-a}$$

$$= g'(f(a)) \cdot f'(a)$$

$$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$$

$$g'(f(a)) = \lim_{\substack{y\to f(a)\\ \updownarrow \\ f(x)}} \frac{g(\overset{\sim f(x)}{y}) - g(f(a))}{\underset{\substack{\updownarrow\\ f(x)}}{y - f(a)}}$$

Q: What is wrong with this proof? ✱ Need to formalize the switch from y to $f(x)$. ✱ What if $f(x) = f(a)$?

Chapter 5: Differentiation

Section 29: The Mean Value Theorem

Motivating Questions

Question: How do we rigorously show the following?

- If $f'(x) = 0$ for all x, then f is constant.
- If $f'(x) > 0$ for all x, then f is strictly increasing.
- If $f'(x) < 0$ for all x, then f is strictly decreasing.

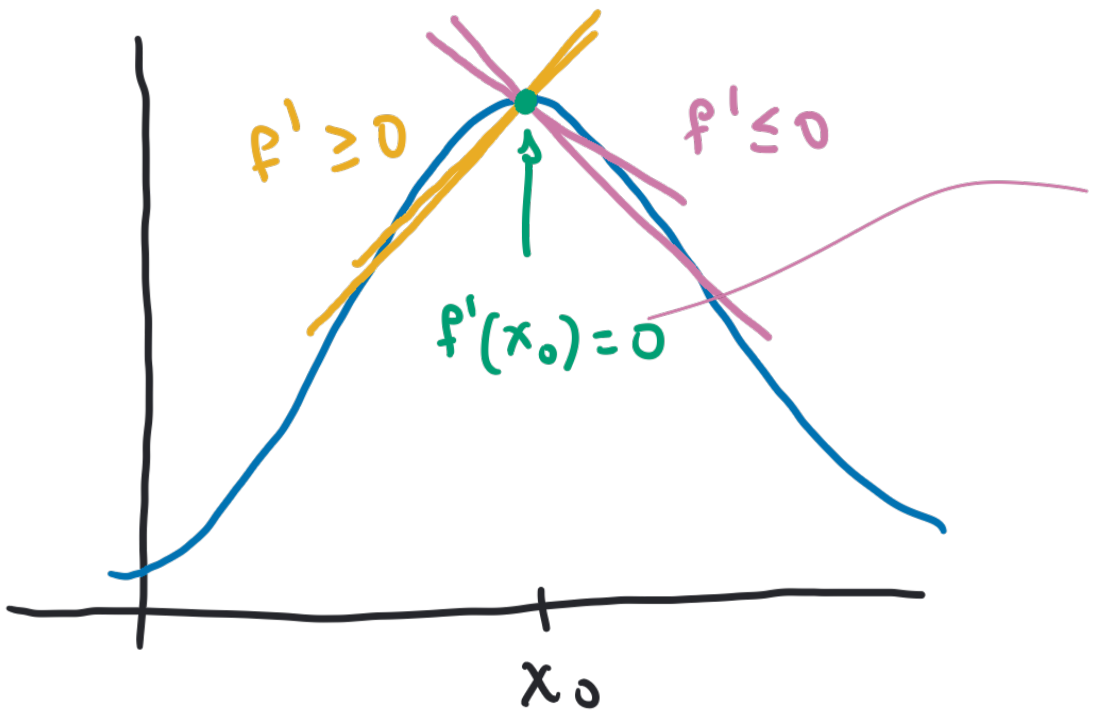
Answer: The Mean Value Theorem!

Extreme Values

neighborhood of x_0

(a,b) $x_0 \in (a,b)$

Theorem 29.1. If f is defined on an open interval containing x_0, if f assumes its maximum or minimum at x_0, and if f is differentiable at x_0, then $f'(x_0) = 0$.



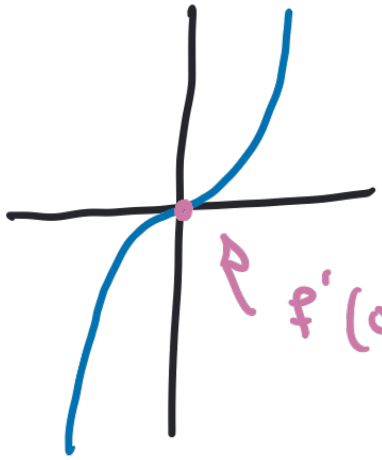
Discussion Questions – True/False

1. If $f'(x_0) = 0$, then x_0 is a maximum or a minimum.

FALSE.

$f(x) = x^3$

$x_0 = 0$.



$f'(0) = 0$

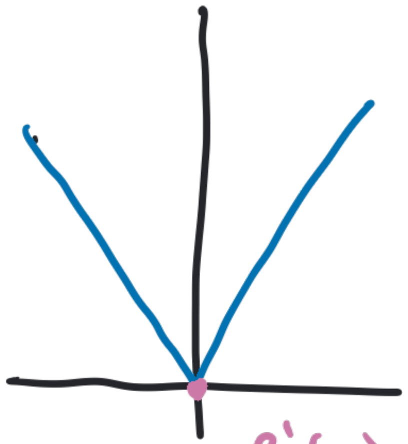
inflection point

2. If x_0 is a maximum or a minimum, then $f'(x_0) = 0$.

FALSE.

$f(x) = |x|$

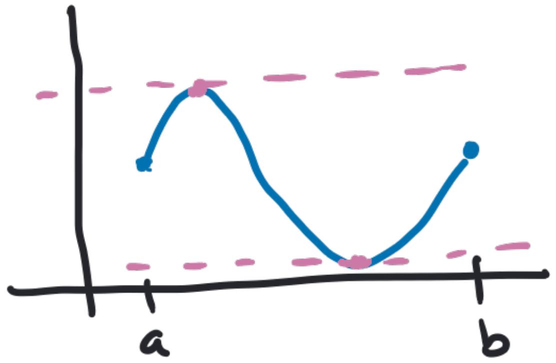
$x_0 = 0$.



cusp

$f'(0)$ does not exist

Rolle's Theorem



Theorem 29.2. Let f be a continuous function on $[a, b]$ that is differentiable on (a, b) and satisfies $f(a) = f(b)$. There exists (at least one) $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Discussion.

- EVT – max and min exist in $[a,b]$
- Theorem 29.1 – if there is max or min in (a,b) then we're done.
- what if max and min are at endpoints?
  ↳ f is constant, ~ $f'(x)=0$

Rolle's Theorem

Theorem 29.2. Let f be a continuous function on $[a, b]$ that is differentiable on (a, b) and satisfies $f(a) = f(b)$. There exists (at least one) $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Proof. Since f is continuous on $[a,b]$, there exist $x_{min}, x_{max} \in [a,b]$ such that $f(x_{min}) \le f(x) \le f(x_{max})$ for all $x \in [a,b]$, by EVT.

Case 1: If $x_{min} \in (a,b)$. Since f is differentiable on (a,b), $f'(x_{min})=0$ by Theorem 29.1.

Case 2: If $x_{max} \in (a,b)$. Since f is differentiable on (a,b), $f'(x_{max})=0$ by Theorem 29.1.

Case 3: If $x_{min}, x_{max} \in \{a,b\}$. Then $f(x_{min}) = f(x_{max})$ since $f(a)=f(b)$. So f is constant. Therefore $f'(x)=0$ for all $x \in (a,b)$.

Mean Value Theorem

Theorem 29.3. If f is continuous on $[a, b]$ and differentiable on (a, b), then there is a number $x_0 \in (a, b)$ such that

$$f'(x_0) = \frac{f(b) - f(a)}{b - a}$$